INDIAN INSTITUTE OF TECHNOLOGY PATNA

EC3202: Digital Signal Processing Lab

EXPERIMENT NO: 02

Linear and Circular Convolution

Submitted by -

Name : Abhitesh Shukla

Roll Number : 2301EE52

Group : 01(Wednesday)

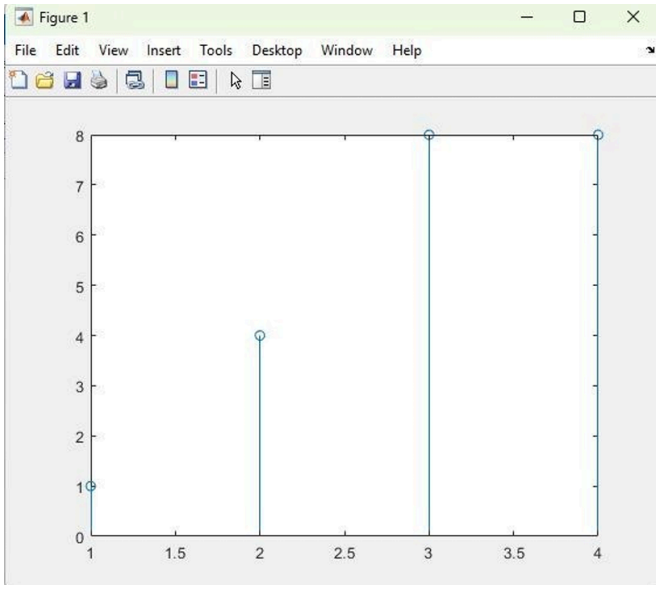
Aim 1:

To compute the response of a discrete LTI system within put sequence x(n) and impulse response h(n) by using linear convolution without using the inbuilt conv function. Verify your program using the conv function for sequences x = [1, 2] and h = [1, 2, 4]

MATLAB Code:

```
x = [1 2];
h = [1 2 4];
n1 = length(x);
n2 = length(h);
n = n1+n2-1;
y = zeros(1, n);
for i = 1:n
	for j = 1:n
		if(i-j+1 > 0 && i*j+1<=n2)
			y(i) = y(i) + x(j)*h(i-j+1);
		end
	end
end
stem(y);
ybuilt_in = conv(x, h);
%stem(ybuilt_in);
```

Plot:



Aim 2:

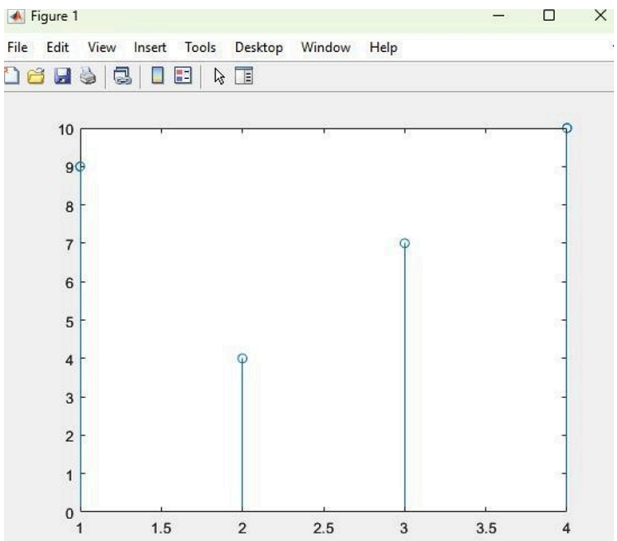
Write a program in MATLAB to compute the response of a discrete LTI system with input sequence x(n) and impulse response h(n) by using Circular convolution. Check your result for the sequences x(n) = [1,2,3,4] and h(n) = [1,2].

MATLAB Code:

```
x2 = [1 2 3 4];
h2 = [1 2];
n12 = length(x2);
n22 = length(h2);
n2 = max(n12, n22);
x2_pad = [x2 , zeros(1, n2 - n12)];
h2_pad = [h2 , zeros(1, n2 - n22)];
y_circ = zeros(1, n2);
for i = 0:n2-1
        sum = 0;
```

```
        for j = 0:n2-1
                h_ind = mod(i-j, n2);
                sum = sum + x2_pad(j+1)*h2_pad(h_ind+1);
        end
        y_circ(i+1) = sum;
end
%stem(y_circ)
```

Plot:



Aim 3:

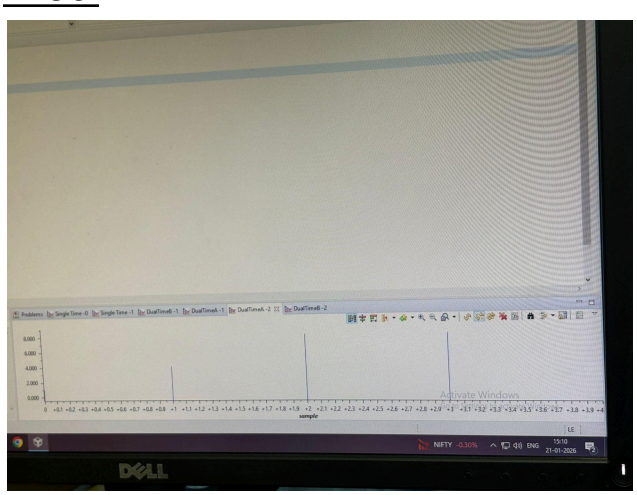
To write a "C" program to verify Linear Convolution and Circular Convolution graphically using Code Composer Studio (CCS-3.1) IDE and the DSP TMS 320C6713 DSK Kit.

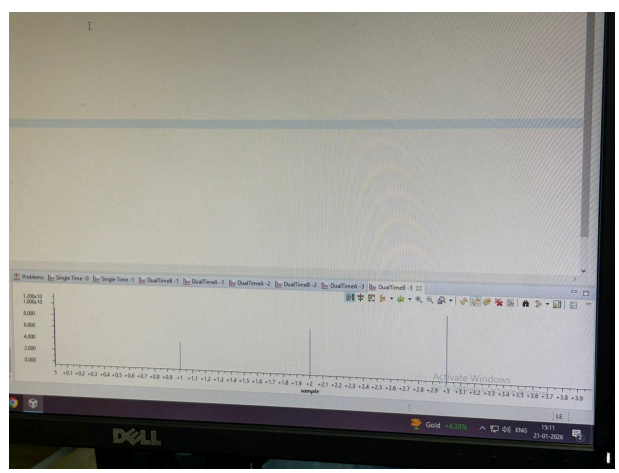
MATLAB Code:

```
#include <stdio.h>
int x[2] = {1, 2};                    //Define Input Sequences
int h[3] = {1, 2, 4};
int y[4] = {0};
```

```
int N1 = 2;                     //Length of x
int N2 = 3;                     //Length of h
int N;                          //Length of y
void main() {
        int i, j;
        N = N1 + N2 - 1;        //Initialize output to 0
        for(i = 0; i < N; i++) {
                y[i] = 0;
                }
        for(i = 0; i < N; i++) {
                for(j = 0; j < N1; j++){
                        if((i - j) >= 0 && (i - j) < N2){
                                y[i] += x[j] * h[i - j];
                        }
                }
        }
// To view this result, you would use the "Watch Window" "Graph"
tool in CCS and observe the array 'y'. // or // while(1); // Infinite
loop to keep program running
}
```

Plot:





Conclusion:

- In this experiment, we analysed the fundamental operation of convolution for discrete-time systems.
- We successfully implemented Linear Convolution manually in MATLAB to determine the response of an LTI system and verified the results using the built-in conv function.
- Additionally, we explored Circular Convolution, which is essential for frequency-domain analysis using the DFT. By matching the lengths of sequences through zero-padding, we computed the circular response and observed how it differs from linear convolution.
- Finally, we implemented these concepts in C code using the TMS320C6713 DSK, bridging the gap between theoretical mathematical models and real-time hardware signal processing.